LAB 5 — Internet Edge + ISP Serial WAN + Week 03 Microservices

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Excellent. We will now **strictly use your new topology and configuration**:

- ✅ R1 G0/0 → Internet (DHCP, NAT outside)
- ✅ R1–R2 connected via **Serial S0/0/0**
- ✅ ISP domain: **100.10.10.0/30**
- ✅ LAN A: 192.168.10.0/24
- ✅ LAN B: 192.168.20.0/24
- ✅ Microservices from **Week03 (Phase 1 Mockup)** FastAPI architecture

We now reframe this as:

🧪 LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

🎯 LAB OBJECTIVE

Transform Week03 microservices into a **realistic enterprise deployment**:

- R1 = Internet Edge Router (NAT + DHCP WAN)
- Serial WAN = ISP Domain (100.10.10.0/30)
- R2 = Branch Router
- Microservices deployed on ServerA and ServerB
- Validate cross-site microservice access over routed WAN + NAT

🗺️ LAB 5 TOPOLOGY (Reframed)

None

```
                        INTERNET
                           |
                       (DHCP)
                        R1 G0/0
                           |
                    [NAT OUTSIDE]
                           |
                    R1 S0/0/0
                 100.10.10.1/30
                           |
                    R2 S0/0/0
                 100.10.10.2/30
                           |
              -----------------------
              |                      |
      R1 G0/1 (LAN A)       R2 G0/1 (LAN B)
      192.168.10.1           192.168.20.1
              |                      |
          ServerA                ServerB
      Week03 Microservices   Week03 Microservices
```

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#description Connect_to_Internet
Router(config-if)#ip address dhcp
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Copy Paste

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES DHCP   up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/3/0            unassigned      YES unset  administratively down down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

```
% Invalid input detected at '^' marker.

Router(config)#hostname R1
R1(config)#interface g0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#interface g0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R1(config)#acess-list 1 permit 192.168.10.0.0.0.0.255
            ^
% Invalid input detected at '^' marker.

R1(config)#access-list 1 permit 192.10.0.0.0.0.255
                                ^
% Invalid input detected at '^' marker.

R1(config)#access-list 1 permit 192.168.10.0.0.0.0.255
                                ^
% Invalid input detected at '^' marker.

R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface g0/1 overload
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#
```

📍 ADDRESS TABLE (FINAL)

Device	Interface	IP	Role
R1	G0/0	DHCP	Internet
R1	S0/3/0	100.10.10.1/30	ISP
R1	G0/1	192.168.10.1/24	LAN A
R2	S0/0/0	100.10.10.2/30	ISP

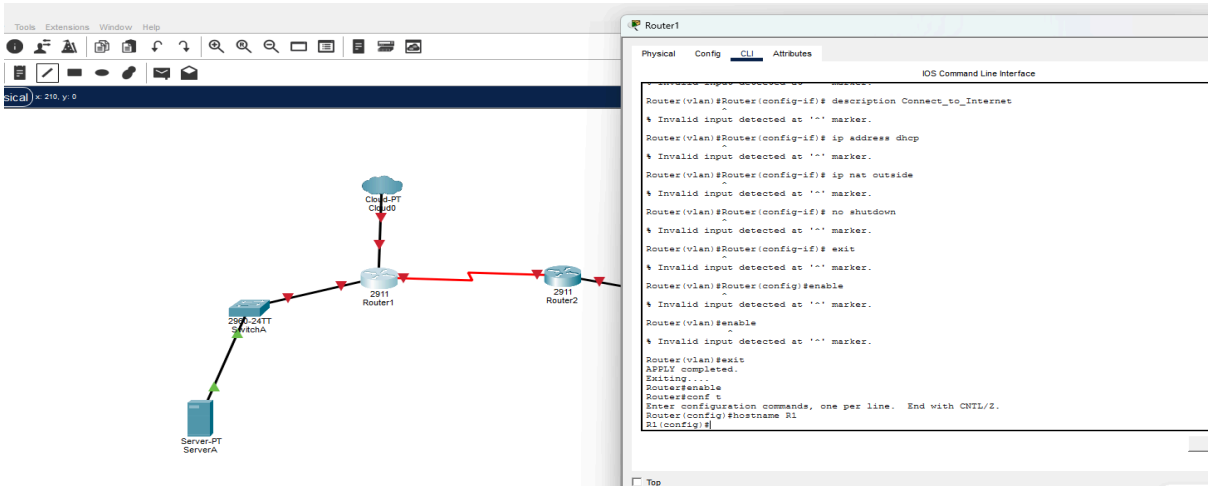
R2	G0/1	192.168.20.1/24	LAN B
ServerA	eth0	192.168.10.10	Microservices A
ServerB	eth0	192.168.20.10	Microservices B

🔧 PART 1 — CISCO CONFIGURATION (STRICT)

🧱 R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/1
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/0
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

interface s0/3/0
 ip address 100.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0
```

```
access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```




```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#description Connect_to_Internet
Router(config-if)#ip address dhcp
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

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```
Router(config)#hostname R1
R1(config)#interface g0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#interface g0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
R1(config)#interface s0/3/0
R1(config-if)#ip address 100.10.10.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
R1(config-if)#exit
R1(config)#
```

```
R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface g0/1 overload
R1(config)#exit
```

```
R1(config-if)#exit
R1(config)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#
```

🧱 R2 — Branch Router

None

```
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/3/0
 ip address 100.10.10.2 255.255.255.252
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1
```

```
end
write memory
```

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#exit
R2(config)#
```

```
R2(config-if)#exit
R2(config)#interface s0/3/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
exit
R2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

```
R2(config)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R2#
```

💻 PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

Services:

- Upload → 8000
- Processing → 8001
- AI → 8002
- Gateway → 9000

ServerA IP:

```none
192.168.10.10
```

🛠️ On ServerB (LAN B)

Same deployment:

```shell
python start_services.py
```

ServerB IP:

```
192.168.20.10
```

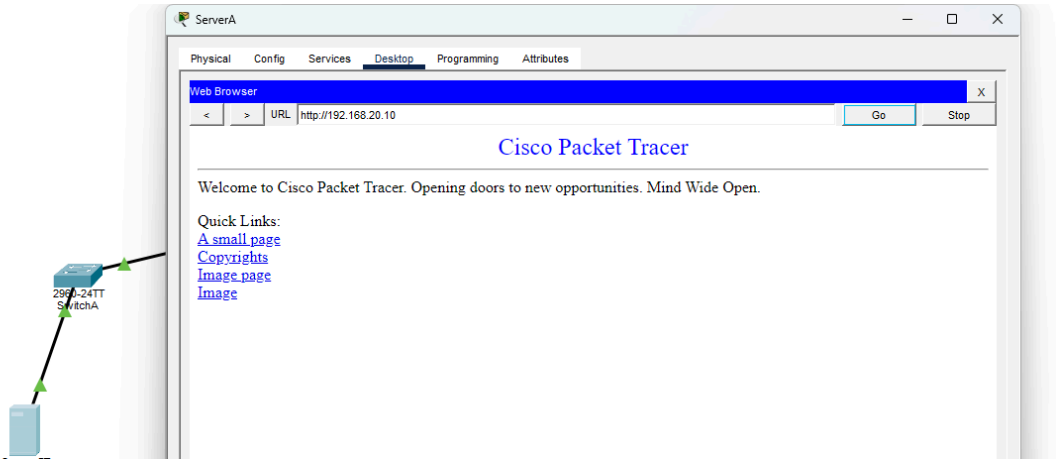
```
🎯 Starting Phase 1 Microservices...
==================================================
🚀 Starting Upload Service on port 8001...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8002...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8003...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 8090...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!
```

```
🔍 Checking system health...
System status: healthy
  ✅ gateway: healthy
  ✅ upload: healthy
  ✅ processing: healthy
  ✅ ai: healthy
```

📊 Service Status:

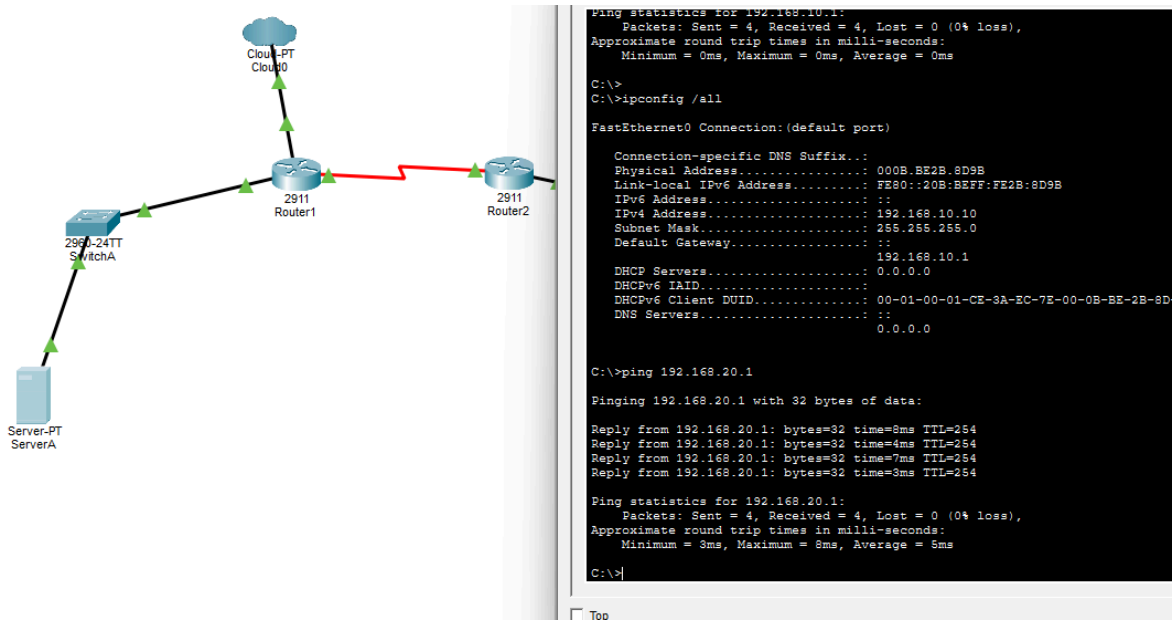
Upload Service 🟢 Running
Processing Service 🟢 Running
AI Service 🟢 Running
Gateway Service 🟢 Running

```
configfile: pytest.ini
plugins: anyio-4.13.0, asyncio-1.3.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 22 items / 21 deselected / 1 selected

tests/test_phase1_compliance.py::TestPhase1Compliance::test_gateway_health
------------------------------------------------------------------------------ live log call ------------------------------------------------------------------------------
INFO     httpx:_client.py:1740 HTTP Request: GET http://localhost:8090/health "HTTP/1.1 200 OK"
PASSED                                                                                                                                                                  [100%]

========================================================================== 1 passed, 21 deselected in 1.66s ==========================================================================
PS C:\network\automation\automation-main\week03-microservices\phase1>
```



การเชื่อมต่อสำเร็จ: ข้อมูลวิ่งข้ามจาก **ServerA (192.168.10.10)** ผ่าน Router R1 และ R2 ไปดึงหน้าเว็บจาก **ServerB (192.168.20.10)**

Static Routing: ตั้งค่าให้ Router ทั้งสองฝั่งรู้จักทางไปหากัน
Distributed Microservices: จำลองการรัน Service ข้าม Site ได้สำเร็จ



A ping 192.168.20.1

🧪 PART 3 — WAN + INTERNET TESTING

🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
INFO     httpx:_client.py:1740 HTTP Request: DELETE http://localhost:8001/upload/241a6772-9ae3-49ac-be69-9dbd1eec24fe "HTTP/1.1 200 OK"
INFO     httpx:_client.py:1740 HTTP Request: GET http://localhost:8001/upload/241a6772-9ae3-49ac-be69-9dbd1eec24fe "HTTP/1.1 404 Not Found"
PASSED                                                                                                                                  [ 50%]
tests/test_phase1_compliance.py::TestPhase1Compliance::test_upload_nonexistent_file
------------------------------------------------------------------------------ live log call ------------------------------------------------------------------------------
INFO     httpx:_client.py:1740 HTTP Request: GET http://localhost:8001/upload/nonexistent_file_123 "HTTP/1.1 404 Not Found"
PASSED                                                                                                                                  [ 62%]
tests/test_phase1_compliance.py::TestPhase1Compliance::test_upload_only_workflow
------------------------------------------------------------------------------ live log call ------------------------------------------------------------------------------
INFO     httpx:_client.py:1740 HTTP Request: POST http://localhost:8090/upload-only "HTTP/1.1 200 OK"
PASSED                                                                                                                                  [ 75%]
tests/test_phase1_compliance.py::TestPhase1Compliance::test_invalid_file_upload
------------------------------------------------------------------------------ live log call ------------------------------------------------------------------------------
INFO     httpx:_client.py:1740 HTTP Request: POST http://localhost:8001/upload "HTTP/1.1 422 Unprocessable Entity"
PASSED                                                                                                                                  [ 87%]
tests/test_phase1_compliance.py::TestPhase1Compliance::test_concurrent_uploads
------------------------------------------------------------------------------ live log call ------------------------------------------------------------------------------
INFO     httpx:_client.py:1740 HTTP Request: POST http://localhost:8001/upload "HTTP/1.1 200 OK"
INFO     httpx:_client.py:1740 HTTP Request: POST http://localhost:8001/upload "HTTP/1.1 200 OK"
INFO     httpx:_client.py:1740 HTTP Request: POST http://localhost:8001/upload "HTTP/1.1 200 OK"
PASSED                                                                                                                                  [100%]

================================================================= 8 passed, 14 deselected in 4.75s ==================================================================
```

```
None
curl http://192.168.20.10:8000/health
```

Should traverse:

ClientA → R1 → Serial ISP → R2 → ServerB

🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl -X POST http://192.168.20.10:9000/process-file
```

Flow:

Upload → Processing → AI → Response returned across WAN

🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```

Verify NAT overload working on R1:

```
None
show ip nat translations
```

🔎 Verification Commands

On R1

```
None
show ip route
show ip nat translations
show ip interface brief
```

On R2

```
None
show ip route
ping 100.10.10.1
```

🧠 Engineering Reflection (Week03 Integration)

From your Week03 profile :

This lab now demonstrates:

Concept	Where Demonstrated
Public vs Private Boundaries	R1 NAT edge
Distributed Services	ServerA & ServerB
Network Segmentation	Serial ISP /30
Horizontal Architecture	Two independent microservice stacks
Internet Edge Pattern	R1 as NAT Gateway
Service Isolation	LAN A vs LAN B

🔬 What Changed from Lab 4

Lab 4	Lab 5
10.10.12.0/30 simulated internet	Real ISP serial 100.10.10.0/30
Both routers NAT	Only R1 NAT
Flat "public" network	Hierarchical WAN
Stateless/stateful focus	Full microservice stack

🏗️ Architectural Significance

You now have:

- Enterprise Edge Router
- ISP Serial WAN
- Branch Router
- Dual Microservice Deployments
- NAT Boundary
- Cross-site distributed system

This is no longer a classroom NAT lab.

This is a **miniature cloud-edge + branch distributed architecture.**